

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech

Branch : Mining Engineering

Semester : 6th Semester

Subject Code & Name: BTMIOE605C Mineral Processing Technology

Max Marks: 60

Date: 30/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The process of crushing and grinding ore into smaller fragment is called _____.	CO-1	1
	a. Mineral processing CO c. Communication d. Reduction		
2	Which of the following is an impact mechanism in Mill?	CO-1	1
	a. Cutter mill CO c. Ball mill d. Fluid energy mill		
3	In a Ball mill how much volume should be occupied by material?	CO-1	1
	a. 50-70% b. 60-80% c. 20-30% d. 40-45%		
4	Mesh denotes the number of opening in a _____.	CO-2	1
	a. Square inch b. Linear Inch c. Square mm d. linear mm		
5	Which of the following screen has a maximum capacity?	CO-2	1
	a. Grizzlies b. Trommels c. Shaking screens d. Vibratory screen		
6	Mechanical classifier is a type of _____.	CO-2	1
	a. Hydro cyclone b. Vertical classifier c. Horizontal current classifier d. Multi spigot hydrosizer		

7	Basic principle of particle separation in gravity concentration process is based on_____.			CO-3	1
	a. difference in particle	b. Settling velocity	c. Size Density	d. None	
8	Which of the following is not a gravity concentrator?			CO-3	1
	a. Crusher	b. Spiral	c. Mill	d. None	
9	Froth floatation is the more suitable for treating_____.			CO-4	1
	a. iron ore	b. Sulphide ore	c. quartzite	d. None	
10	The oil used as a frothing agent in froth floatation process is_____.			CO-4	1
	a. Coconut oil	b. Castore oil	c. Palm oil	d. Pine Oil	
11	Washing Of Coal_____.			CO-5	1
	a. Reduces its sulphur and ash content	b. Control its ash fusibility and increases its calorific value	c. Improves its cocking properties	d. All a,b,c	
12	Today the sedimentation equipment widely known as _____are manufacturing in many designs and shape.			CO-5	1
	a. Thickeners	b. Filters	c. Centrifugal filters	d. Electro ferric filters	
Q. 2	Solve the following.				12
A)	Describe the working of double toggle blake jaw crusher.			CO-1	6
B)	Discuss the theory of ball mill operation with respect to the following: 1. Various zones in the mill 2. Effect of the speed of the mill			CO-1	6
Q.3	Solve the following.				12
A)	Explain the term particle size distribution giving suitable examples.			CO-2	6
B)	Describe with neat sketch hydro-cyclone.			CO-2	6

Q. 4	Solve Any Two of the following.		12
A)	Describe the construction of Wilfley table.	CO-3	6
B)	Discuss the principles of sink and float giving suitable flowsheet.	CO-3	6
C)	Explain the Bio-Leaching.	CO-3	6
Q.5	Solve Any Two of the following.		12
A)	Explain with neat sketch the working of floatation column.	CO-4	6
B)	What is froth floatation? State the function of collector in froth floatation operation.	CO-4	6
C)	Write an essay on magnetic separation in mineral processing giving suitable example.	CO-4	6
Q. 6	Solve Any Two of the following.		12
A)	What do you understand by washability curves? How they are established?	CO-5	6
B)	What is Pelletization? Why it is carried out? Explain the process of pelletization.	CO-5	6
C)	Describe the application of thickeners in mineral processing plant for water reclamation.	CO-5	6
	*** End ***		